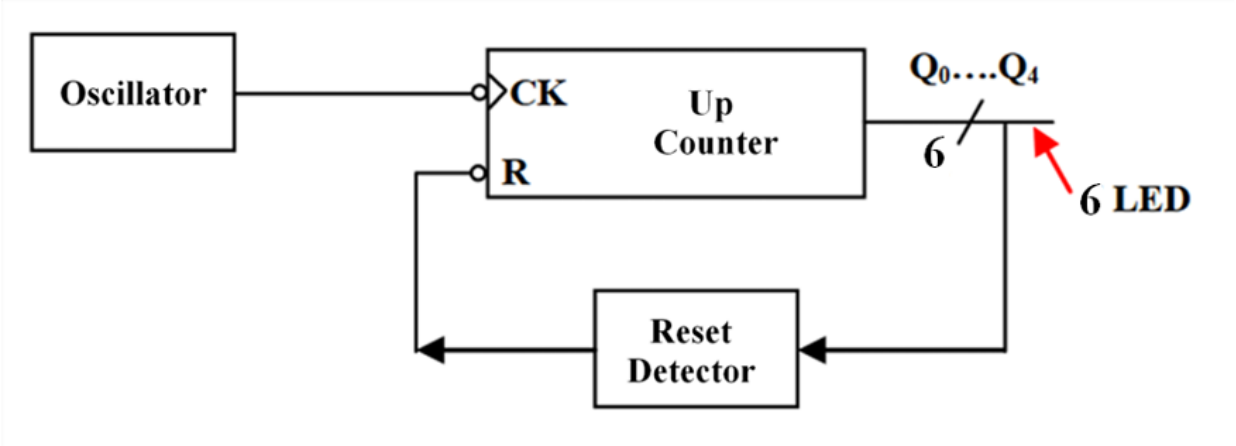
# Counter/divider by 44 with counting sequence 0, 1, 2, …, 42, 43, 0

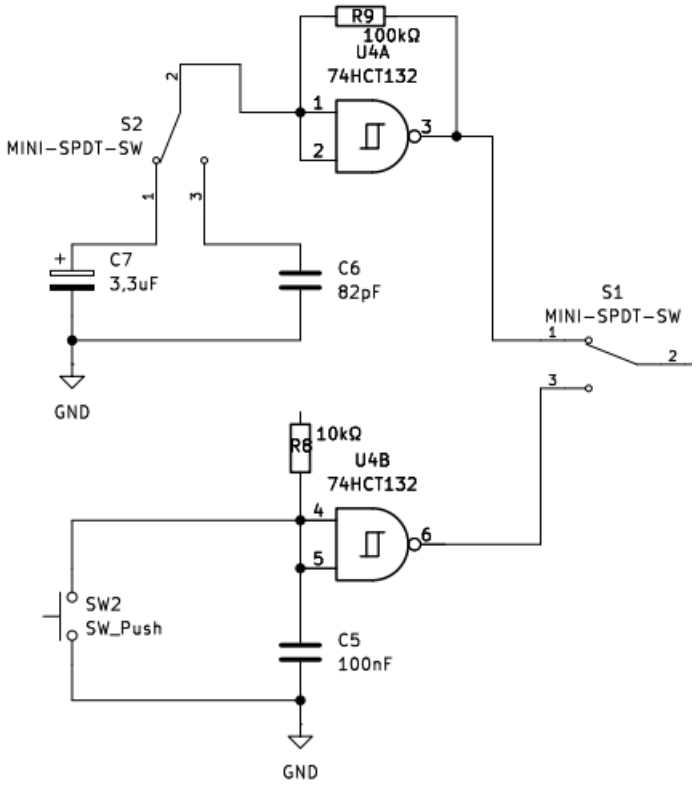
**Students:** Luchian Mara, Mateșescu Corina

**Coordinator:** Papazian Petru

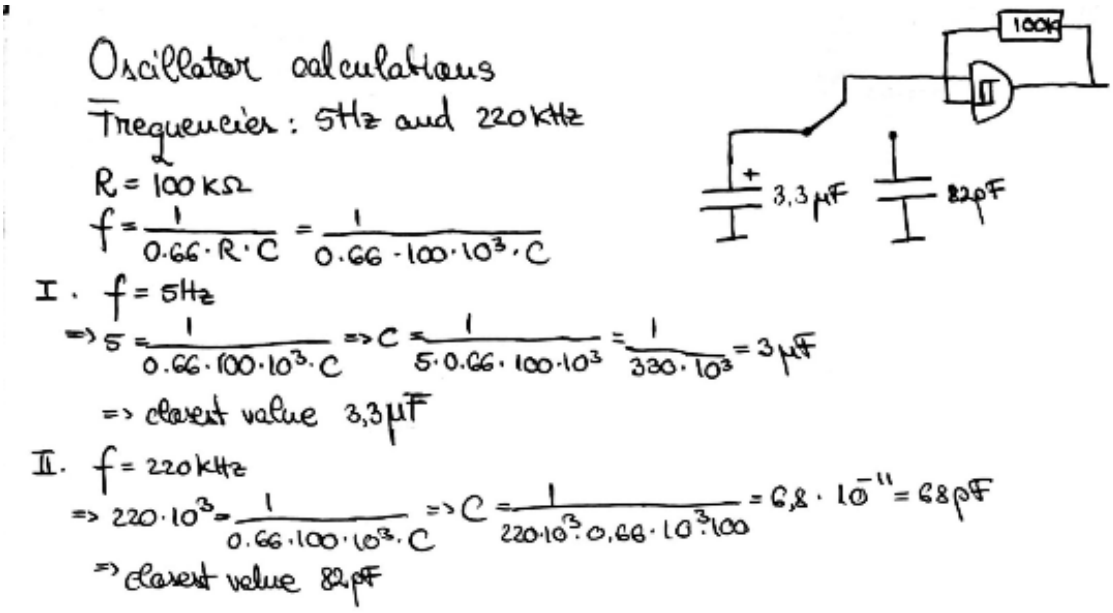
**Counter/divider by 44 with counting sequence 0,1,2…42,43,0. Counter will use the synchronous R input.** It will use only ICs HCT/HC series. Oscillator will have two frequencies of 5Hz, respectively of 220kHz and a manual tick button.



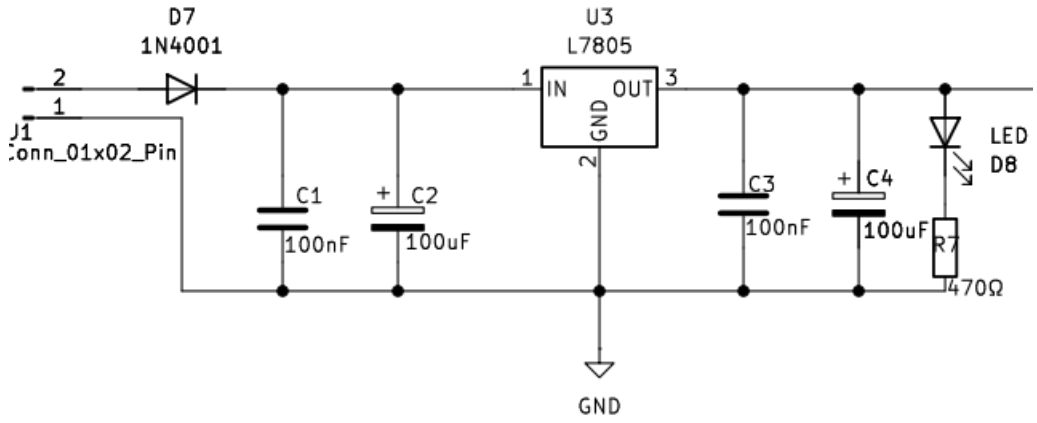

**Block diagram**



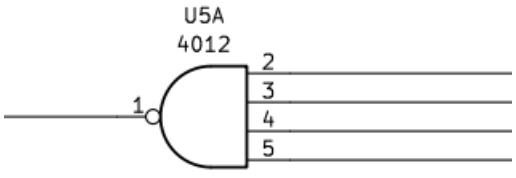

**Oscillator schematic**



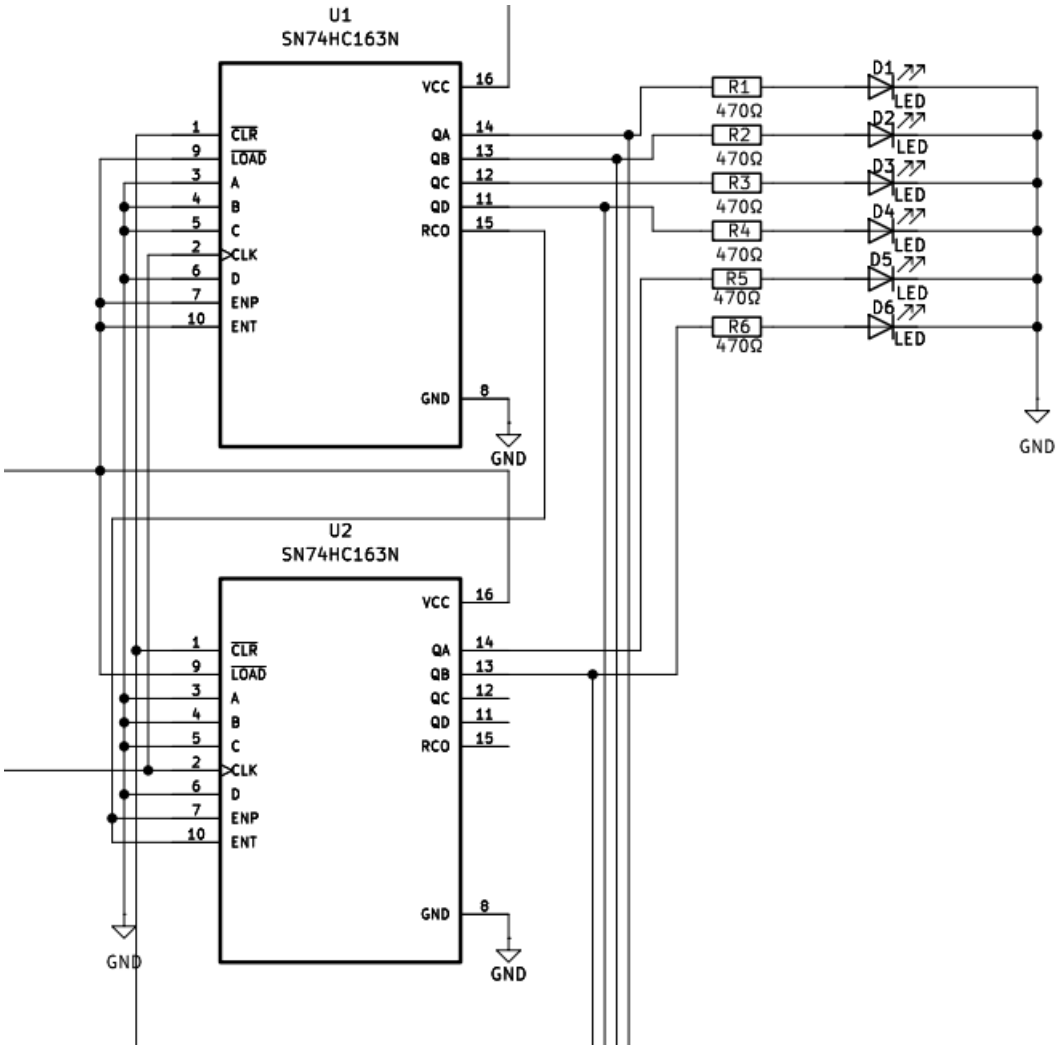

**Oscillator frequencies calculations**



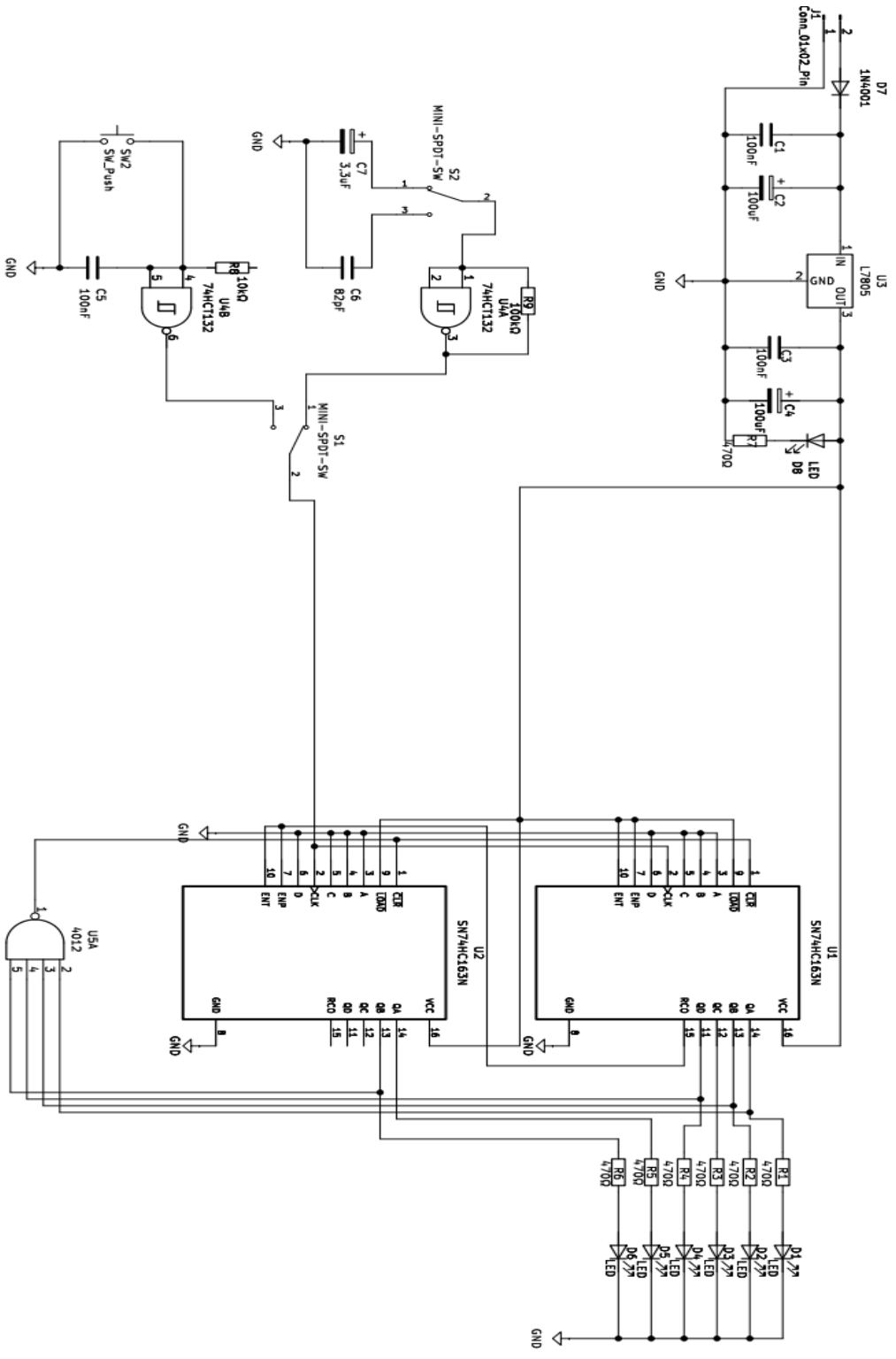

**Power supply schematic**



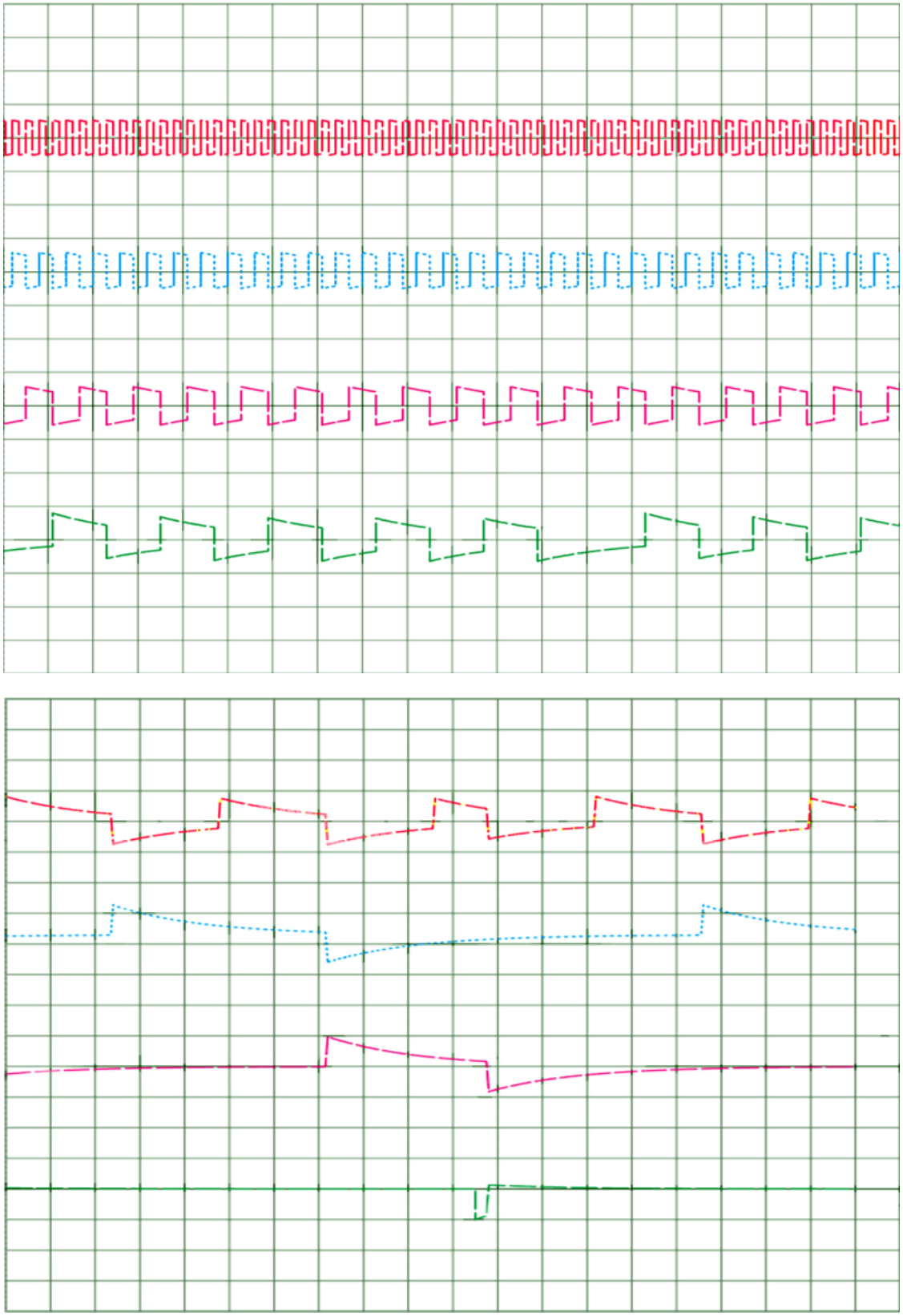

**Reset Detector schematic**



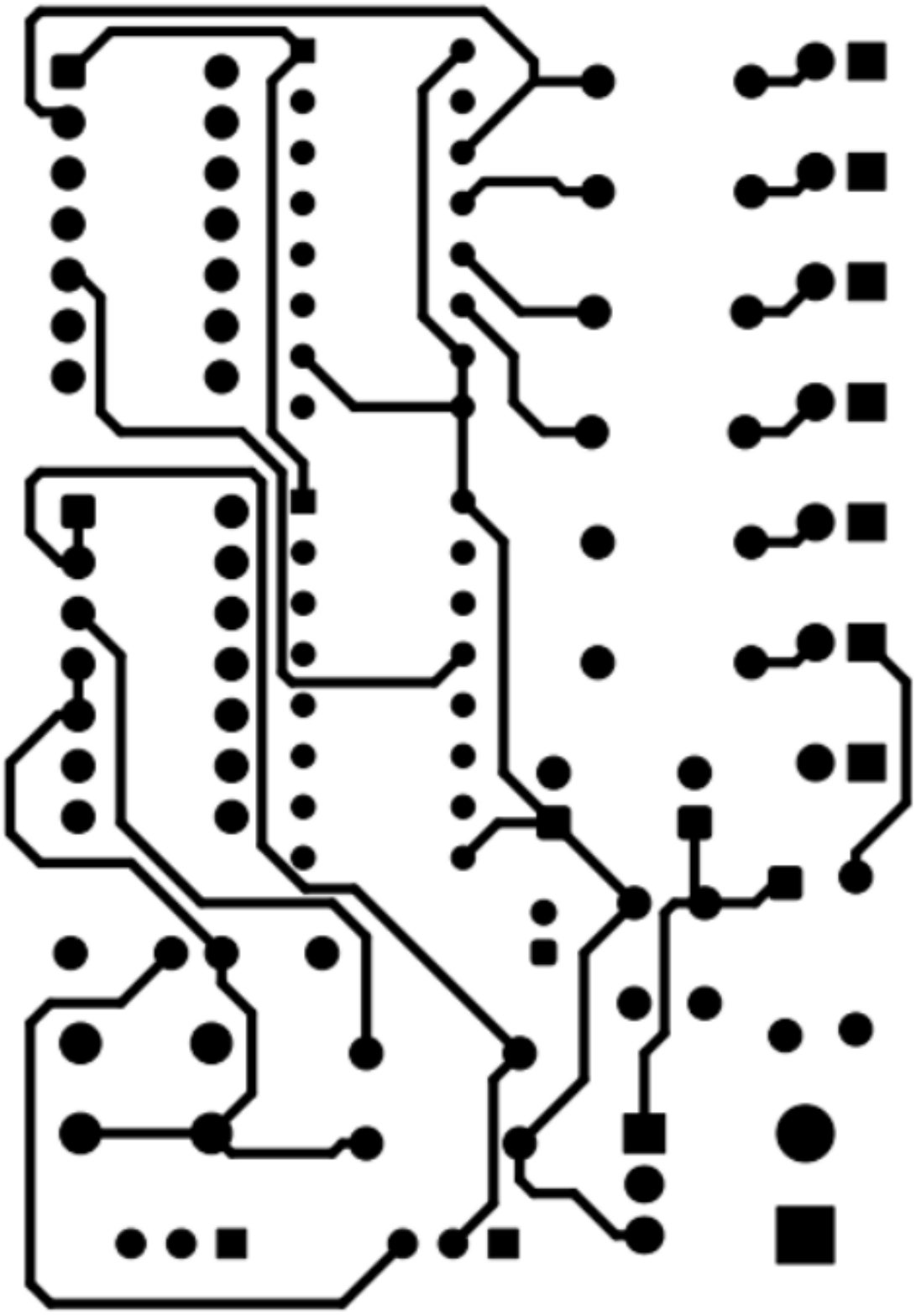

**Counter schematic**



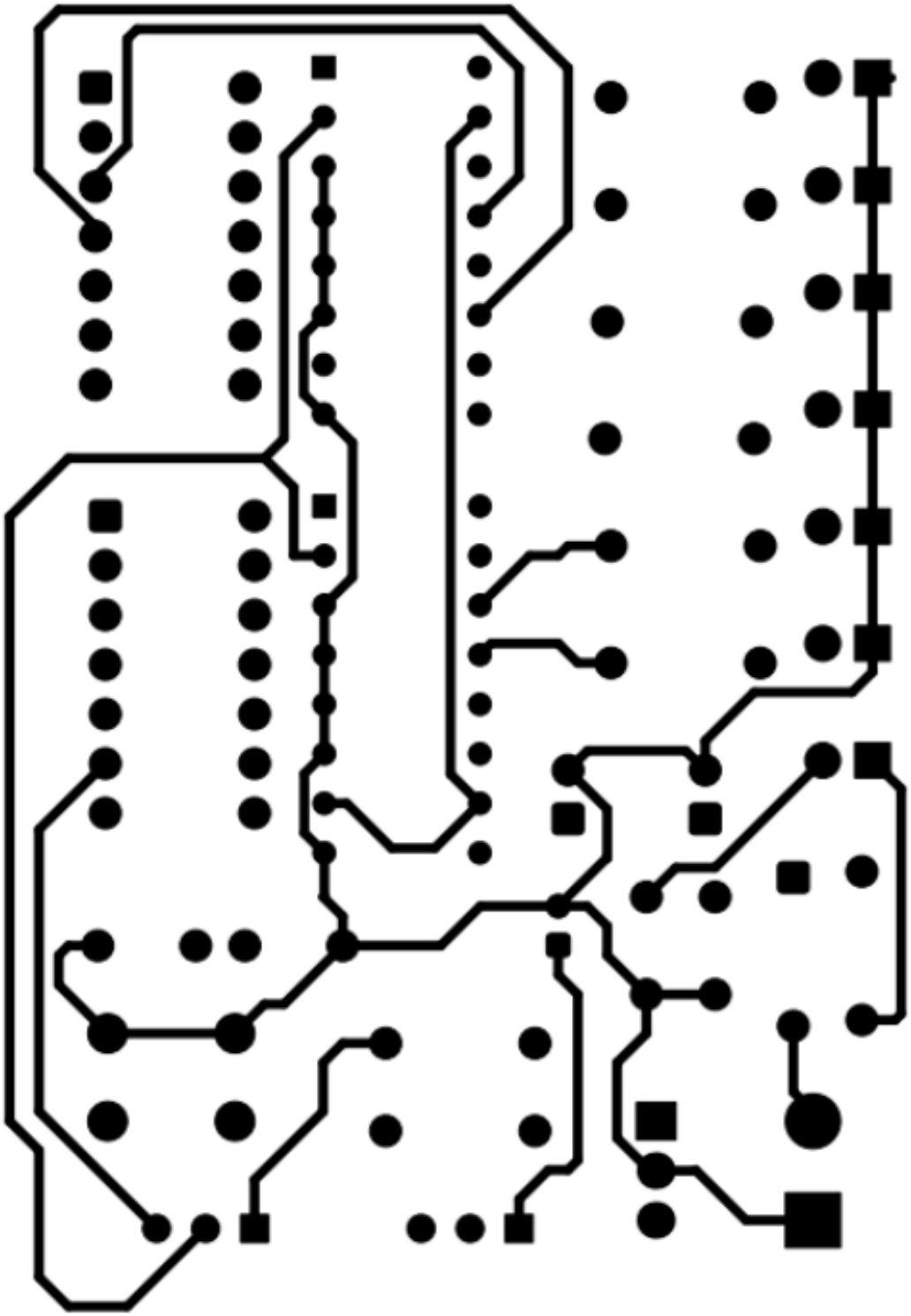

**Full circuit schematic**



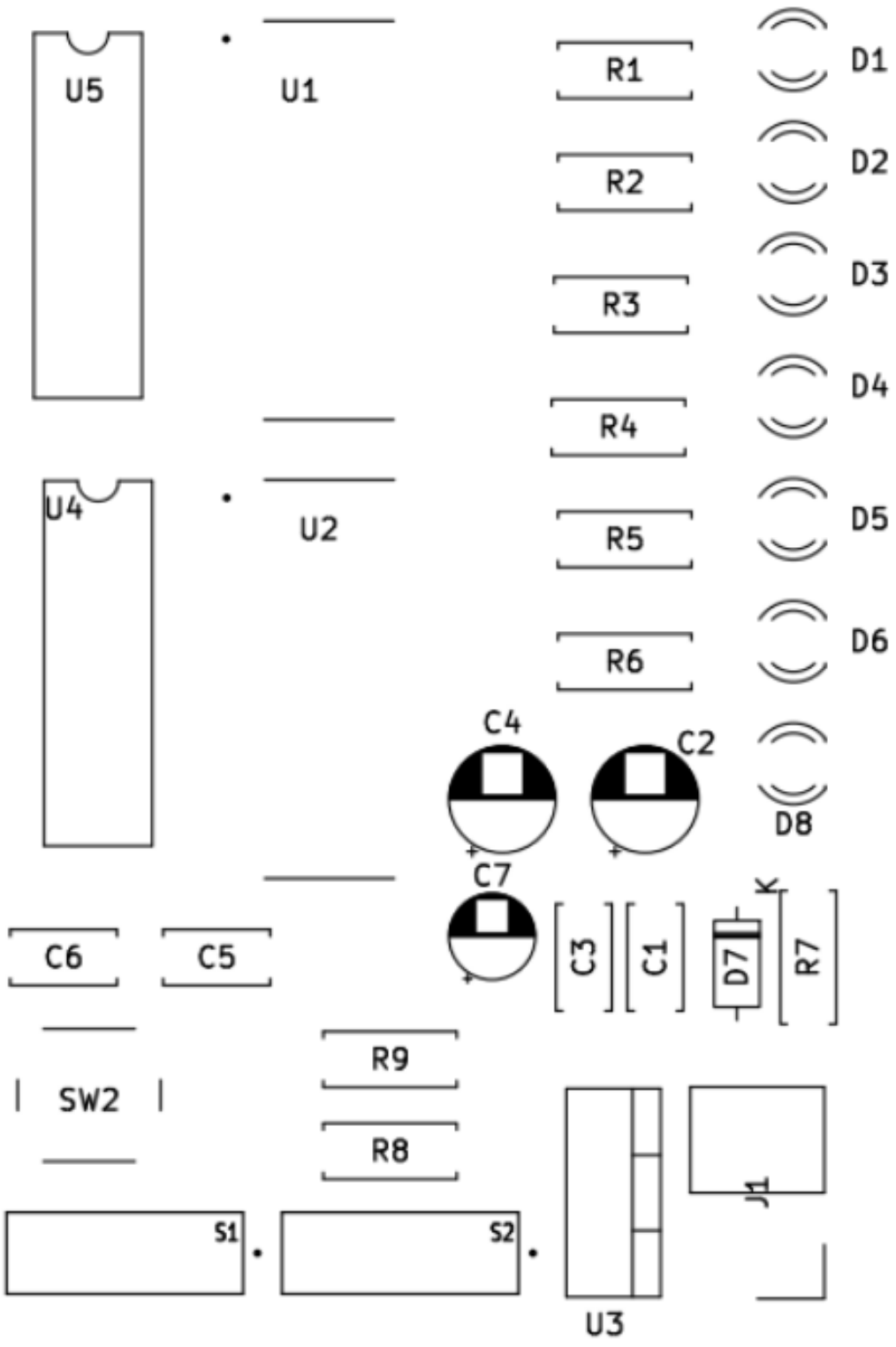
**Simulation graphs**



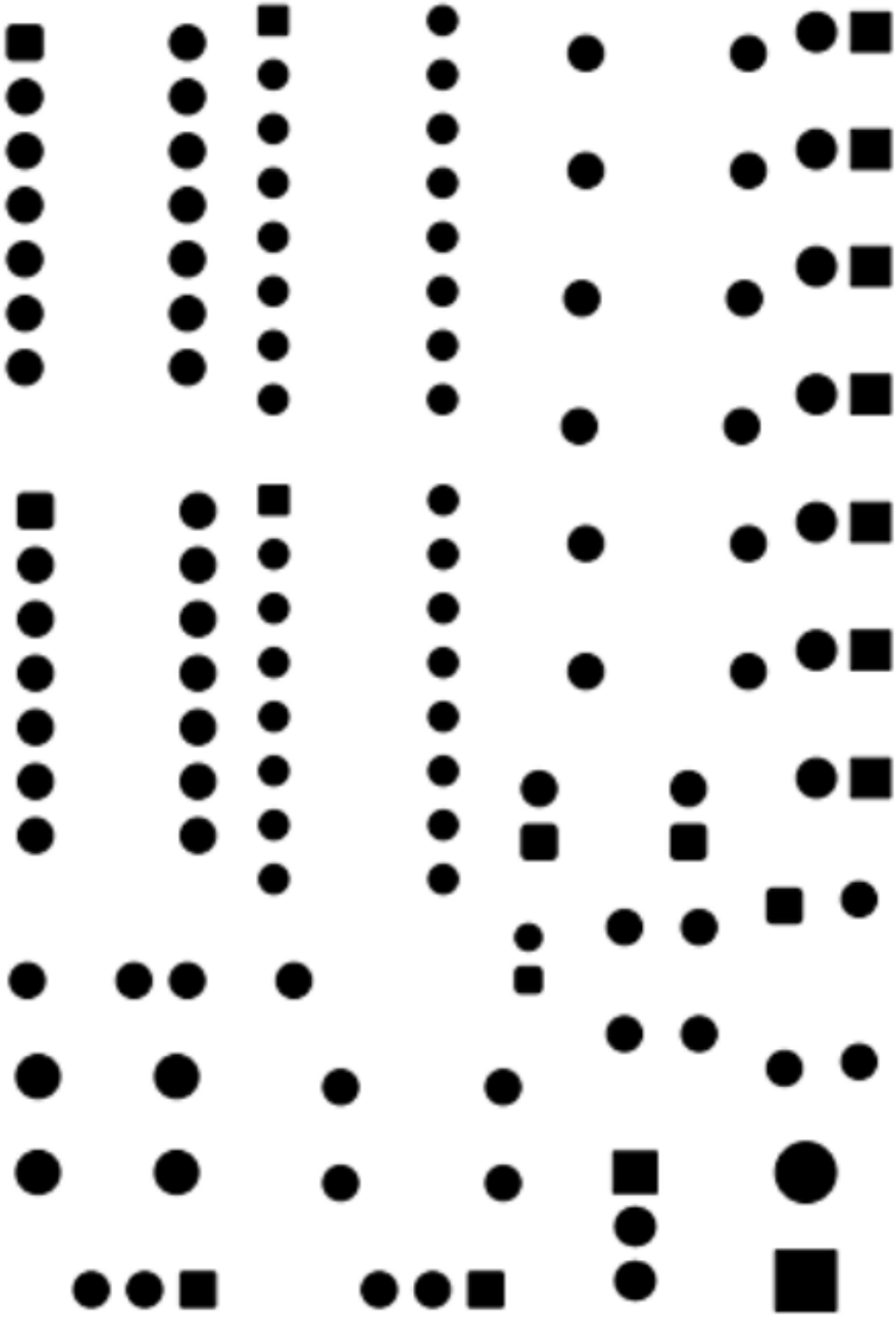
**Top copper layer**



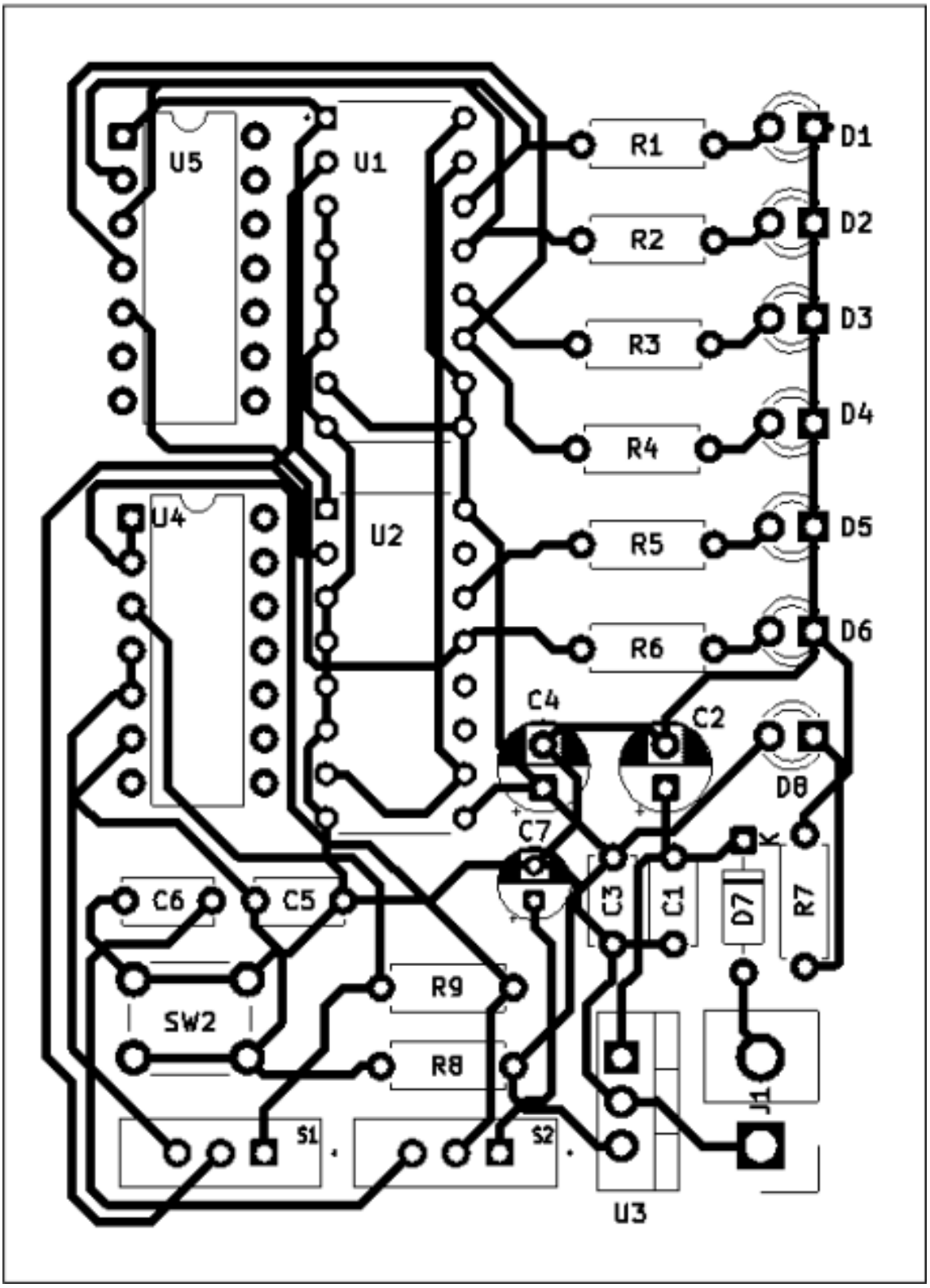
**Bottom copper layer**


U5
U1
R1
D1
R2
D2
R3
D3
R4
D4
U4
U2
R5
D5
R6
D6
C4
C2
D8
C7
K
C6
C5
C3
C1
D7
R7
R9
SW2
R8
J1
S1
S2
U3

**Silkscreen**



**Top and bottom mask**




**PCB transparency**

Power consumption calculations

- 2 x 74HC1G3

  $V_{CC} = 5V$

  $I_{CC_S} = 160\mu A$ $\Big\}$ $\Rightarrow I_{CC} = 50.16mA$ $\Big\}$ $\Rightarrow P_D = 250.8mW$

  $I_L = 50mA$

  $\Rightarrow P_{DT} = 501.6mW$ for both ICs

- 74HCT132

  $V_{CC} = 5V$

  $I_{CC_S} = 40\mu A$ $\Big\}$ $\Rightarrow P_D = 5 \cdot (40\mu A + 50mA) \Rightarrow P_D = 250.2mW$

  $I_{CC} = 50mA$

- LEDs + Resistors

  $I = \frac{V_{CC} - V_{LED}}{R} = \frac{5-3}{470} = \frac{2V}{470\Omega} = 4.26mA$

  $P_R = V_{CC} \cdot I = 5 \cdot 4.26 = 21.28mW$ $\Big\}$ $\Rightarrow$ for one pair $P = 34.04mW$

  $P_{LED} = V_{LED} \cdot I = 3 \cdot 4.26 = 12.77mW$ $\Big\}$

  for 7 pairs $\Rightarrow 7 \cdot 34.04 \Rightarrow P_D = 238.28mW$

- R - oscillator resistor

  $P = U \cdot I$ $\Big\}$ $\Rightarrow P = \frac{U^2}{R}$ $\Rightarrow P = \frac{3.5^2}{100K} = 0.1mW$

  $I = \frac{U}{R}$ $\Big\}$

- CD4012

  $V = 5V$

  $I_S = 15\mu A$ $\Big\}$ $\Rightarrow I = 50.015mA$ $\Big\}$ $\Rightarrow P_D = 5 \cdot 50.015 = 250.08mW$

  $I_L = 50mA$ $\Big\}$

- L7805

  $V = 8.25V$ $\Big\}$ $\Rightarrow P_D = 8.25 \cdot 8.5 = 70mW$

  $I = 8.5mA$ $\Big\}$

$\Rightarrow P_{DT} = 501.6mW + 250.2mW + 238.28mW + 0.1mW + 250.08mW + 70mW$

$P_{DT} = 1.24W$

**Power consumption calculations**

# Bill of materials

| Reference | Qty | Value | Footprint |
|---|---|---|---|
| C1,C3,C5 | 3 | 100nF | Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm |
| C2,C4 | 2 | 100uF | Capacitor_THT:CP_Radial_D5.0mm_P2.50mm |
| C6 | 1 | 82pF | Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm |
| C7 | 1 | 3,3uF | Capacitor_THT:CP_Radial_D4.0mm_P2.00mm |
| D1,D2,D3,D4,D5,D6,D8 | 7 | LED | LED_THT:LED_D3.0mm |
| D7 | 1 | 1N4001 | Diode_THT:D_DO-35_SOD27_P7.62mm_Horizontal |
| J1 | 1 | Conn_01x02_Pin | Connector_Samtec_HPM_THT:Samtec_HPM-02-05-x-S_Straight_1x02_Pitch5.08mm |
| R1,R2,R3,R4,R5,R6 | 6 | 470Ω | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal |
| R7 | 1 | 470Ω | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal |
| R8 | 1 | 10kΩ | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal |
| R9 | 1 | 100kΩ | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal |
| S1,S2 | 2 | MINI-SPDT-SW | Custom_Footprints:SW_MINI-SPDT-SW |
| SW2 | 1 | SW_Push | Button_Switch_THT:SW_PUSH_6mm_H4.3mm |
| U1,U2 | 2 | SN74HC163N | Custom_Footprints:DIP794W45P254L1969H508Q16 |
| U3 | 1 | L7805 | Package_TO_SOT_THT:TO-220-3_Vertical |
| U4 | 1 | 74HCT132 | Package_DIP:DIP-14_W7.62mm |
| U5 | 1 | 4012 | Package_DIP:DIP-14_W7.62mm |